Group Cohomology

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1 Definitions and resolutions

Let G be a group.

Definition 1.1 (integral group ring)
The *integral group ring* $\mathbb{Z}G$ has elements being formal sums

$$\sum n_g g$$

for $n_g \in \mathbb{Z}, g \in G$, and only finitely many n_g are non-zero. This is a free abelian group under addition, i.e.

$$\left(\sum m_g g\right) + \left(\sum n_g g\right) = \sum (m_g + n_g) g$$

and we have multiplication

$$\left(\sum m_h h\right)\left(\sum n_k k\right) = \sum_g \left(\sum_{hk=g} m_h n_k\right) g$$

This is the associative ring underlying the integral representation theory of G. We will write $1 = 1e$ for the multiplicative identity in G.

Definition 1.2 (module)
A *(left) $\mathbb{Z}G$-module* M is an abelian group under addition, and a map

$$\begin{aligned} \mathbb{Z}G \times M &\to M \\ (r, m) &\mapsto rm \end{aligned}$$

such that

1. $r(m+n) = rm + rn$,
2. $(r_1 + r_2)m = r_1 m + r_2 m$,
3. $(r_1 r_2)m = r_1(r_2 m)$,
4. $1m = m$

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A module is *trivial* if $gm = m$ for all $g \in G$, $m \in M$. We say *the trivial module* is $\mathbb{Z}$ with trivial action $gm = m$. A *free $\mathbb{Z}G$-module* on X is formal sums

$$\sum r_x x$$

for $r_x \in \mathbb{Z}G$, $x \in X$, and finitely many r_x non-zero. We will write $\mathbb{Z}G\{X\}$ for this.

Submodules and *quotient modules* are defined as we would expect.

Definition 1.3 ($\mathbb{Z}G$-map)

An *$\mathbb{Z}G$-map (or morphism)* $\alpha : M_1 \to M_2$ is a map of abelian groups, with

$$\alpha(rm_1) = r\alpha(m_1)$$

for $r \in \mathbb{Z}G$, $m_1 \in M_1$.

Example 1.4

The *augmentation map* is

$$\varepsilon : \mathbb{Z}G \to \mathbb{Z}$$
$$\sum n_g g \mapsto \sum n_g$$

where we treat $\mathbb{Z}G$ as a left $\mathbb{Z}G$-module, and $\mathbb{Z}$ as the trivial module. This is a $\mathbb{Z}G$ map, and it is also a ring homomorphism.

Notation 1.5. Write $\mathrm{Hom}_G(M, N)$ for the set of $\mathbb{Z}G$-maps from $M \to N$, with pointwise addition.

Example 1.6

If we regard $\mathbb{Z}G$ as a left $\mathbb{Z}G$-module, then

$$\mathrm{Hom}_G(\mathbb{Z}G, M) \cong M$$

for any left $\mathbb{Z}G$-module M, by sending φ to $\varphi(1)$. This is an isomorphism since

$$\varphi(r) = \varphi(r1) = r\varphi(1)$$

Note that $\mathrm{Hom}_G(\mathbb{Z}G, M)$ is a left $\mathbb{Z}G$-module, with

$$(s\phi)(r) = \phi(rs)$$

In particular,

$$\mathrm{Hom}_G(\mathbb{Z}G, \mathbb{Z}G) \cong \mathbb{Z}G$$

With this, $\varphi \mapsto \varphi(1)$ corresponds to multiplication on the *right* by $\varphi(1)$.

Note that G may not be abelian, so the distinction between left and right modules matters.

Definition 1.7

If $f : M_1 \to M_2$ is a $\mathbb{Z}G$-map, then the dual map is

$$f^* : \mathrm{Hom}_G(M_2, N) \to \mathrm{Hom}_G(M_1, N)$$

for any $\mathbb{Z}G$-module N. This is given by

$$f^*(\varphi) = \varphi \circ f$$

Similarly, if $f : N_1 \to N_2$, then we have

$$f_* : \mathrm{Hom}_G(M, N_1) \to \mathrm{Hom}(M, N_2)$$

with $f_*(\varphi) = f \circ \varphi$.

These are maps of abelian groups, since in general $\mathrm{Hom}_G(M, N)$ doesn't have to be a $\mathbb{Z}G$-module, as we don't have a right action.

Example 1.8 (Prototypical example)

Let $G = \langle t \rangle$ be an infinite cyclic group, and consider the graph Γ with vertices $\{v_i\}_{i\in\mathbb{Z}}$, edges $v_i \leftrightarrow v_{i+1}$. Then G acts on the set V of vertices, by $tv_i = v_{i+1}$, and also on the set E of edges. The action is transitive in both cases. The formal integral sums $\mathbb{Z}V$ and $\mathbb{Z}E$ are $\mathbb{Z}G$-modules, and are free. Fix the edge $e : v_0 \leftrightarrow v_1$. Then we have $\mathbb{Z}G$-maps

$$\begin{aligned} \mathrm{d} : \mathbb{Z}E &\to \mathbb{Z}V \\ e &\mapsto v_1 - v_0 \end{aligned}$$

and also

$$\begin{aligned} \mathbb{Z}V &\to \mathbb{Z} \\ v_0 &\mapsto 1 \end{aligned}$$

This correspond to the augmentation map.

Definition 1.9 (chain complex, exact, homology)

A *chain complex* of $\mathbb{Z}G$-modules is a sequence

$$M_s \xrightarrow{\mathrm{d}_s} M_{s-1} \xrightarrow{\mathrm{d}_{s-1}} \cdots \xrightarrow{\mathrm{d}_{t+1}} M_t$$

of $\mathbb{Z}G$-modules and maps, such that for $t < n < s$, $\mathrm{d}_n \circ \mathrm{d}_{n+1} = 0$. We will write

$$M_\bullet = (M_n, \mathrm{d}_n)_{t\le n\le s}$$

We say that $M_\bullet$ is *exact* at M_n if $\mathrm{im}(\mathrm{d}_{n+1}) = \ker(\mathrm{d}n)$. The sequence is exact if if it exact at all M_n with $t < n < s$.

The *homology* of $M_\bullet$ is given by

$$\begin{aligned} \mathrm{H}_s(M_\bullet) &= \ker(\mathrm{d}s) \\ \mathrm{H}_n(M_\bullet) &= \frac{\ker(\mathrm{d}_n)}{\mathrm{im}(\mathrm{d}_{n+1})} \text{ for } t < n < s \\ \mathrm{H}_t(M_\bullet) &= \mathrm{coker}(\mathrm{d}_{t+1}) \end{aligned}$$

A *short exact sequence* is a chain complex of the form

$$0 \longrightarrow M_1 \xrightarrow{\alpha} M_2 \xrightarrow{\beta} M_3 \longrightarrow 0$$

In our prototypical example, we have a short exact sequence

$$0 \longrightarrow \mathbb{Z}E \longrightarrow \mathbb{Z}V \longrightarrow \mathbb{Z} \longrightarrow 0$$

which corresponds to

$$0 \longrightarrow \mathbb{Z}G \longrightarrow \mathbb{Z}G \longrightarrow \mathbb{Z} \longrightarrow 0$$

where the map $\mathbb{Z}G \to \mathbb{Z}G$ is multiplication on the right by $t - 1$.

Definition 1.10 (projective)

A $\mathbb{Z}G$-module P is *projective* if for every surjective $\mathbb{Z}G$-map $\alpha : M_1 \to M_2$, and every $\mathbb{Z}G$-map $\beta : P \to M_2$, there exists $\overline{\beta}$ such that

$$\begin{array}{ccccc} & & & & P \\ & & \overset{\overline{\beta}}{\nearrow} & & \downarrow{\scriptstyle \beta} \\ M_1 & \xrightarrow[\alpha]{} & M_2 & \longrightarrow & 0 \end{array}$$

commutes.

If we have any short exact sequence

$$0 \longrightarrow N \xrightarrow{f} M_1 \xrightarrow{\alpha} M_2 \longrightarrow 0$$

Consider

$$0 \longrightarrow \mathrm{Hom}_G(P, N) \xrightarrow{f_*} \mathrm{Hom}_G(P, M_1) \xrightarrow{\alpha_*} \mathrm{Hom}_G(P, M_2) \longrightarrow 0 \qquad (*)$$

Then P is projective if and only if $(*)$ is exact. Note that in $(*)$, we always have exactness except at $\mathrm{Hom}_G(P, M_2)$.

Lecture 2

Lemma 1.11. Free modules are projective.

Proof. Let $\alpha : M_1 \to M_2$ be a surjective $\mathbb{Z}G$-map, $\beta : \mathbb{Z}G\{X\} \to M_2$. For each $x \in X$, since α is surjective, there exists $m_x \in M_1$ such that $\alpha(m_x) = \beta(x)$. We can then define $\overline{\beta} : \mathbb{Z}G\{X\} \to M_1$, by

$$\sum r_x \cdot x \mapsto \sum r_x \cdot m_x$$

□

Definition 1.12

A *projective (resp. free) resolution of the trivial module* $\mathbb{Z}$ is an exact sequence

$$\cdots \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

with all the P_i projective (resp. free).

Example 1.13

When $G = \langle t \rangle$ is infinite cyclic, we have the free resolution

$$0 \longrightarrow \mathbb{Z}G \xrightarrow{\cdot(t-1)} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

Example 1.14

If $G = \langle t \rangle$ cyclic of order n. Then

$$\cdots \longrightarrow \mathbb{Z}G \xrightarrow{\alpha} \mathbb{Z}G \xrightarrow{\beta} \mathbb{Z}G \xrightarrow{\alpha} \mathbb{Z}G \xrightarrow{\varepsilon} \mathbb{Z} \longrightarrow 0$$

is a free resolution, where $\alpha(x) = x(t-1)$ and $\beta(x) = x(1 + t + \cdots + t^{n-1})$.

From algebraic topology, if we have a connected simplicial complex X, with $\pi_1(X) = G$, and the universal cover $\widetilde{X}$ is contractible, then we have a free resolution of $\mathbb{Z}$. The point is that the simplicial complex X contains a lot of information about G. We are trying to replicate this algebraically.

For calculation purposes, we're interested in *small resolutions*. For example, where the free modules have small rank. But for theory development purposes, we're wanting general constructions, and such resolutions tend to be large.

Definition 1.15

G is of type FP_n if $\mathbb{Z}$ has a projective resolution

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

so that $P_n, \ldots, P_0$ are finitely generated.

G is of type FP_n is $\mathbb{Z}$ has a projective resolution with all P_n finitely generated.

G is of type FP if there exists a projective resolution of $\mathbb{Z}$ with finite length and all P_n are finitely generated. That is, $P_s = 0$ for s sufficiently large.

Example 1.16

$G = \langle t \rangle$ infinite cyclic is of type FP, $G = C_n = \langle t \rangle$ finite cyclic is of type FP_∞.

These are to be regarded as finiteness conditions on the group. The topological version of FP_n (which is called F_n) would be asking for X with $\pi_1(X) = G$, having a finite n-skeleton. Note that these two conditions are not equivalent.

Now lets meet some general constructions. If we have a partial projective resolution

$$P_s \xrightarrow{d_s} P_{s-1} \longrightarrow \cdots \longrightarrow P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

Then we can make it longer by setting

$$P_{s+1} = \mathbb{Z}G\{X_{s+1}\}$$

with $X_{s+1} = \ker(d_s)$, and set

$$d_{s+1} : \sum r_x x \mapsto \sum r_x x$$

Then

$$P_{s+1} \xrightarrow{d_{s+1}} P_s \xrightarrow{d_s} P_{s-1} \longrightarrow \cdots \longrightarrow P_1 \xrightarrow{d_1} P_0 \xrightarrow{d_0} \mathbb{Z} \longrightarrow 0$$

is a longer partial projective resolution. To make P_{s+1} smaller, we could take X_{s+1} to be a $\mathbb{Z}G$-generating set for $\ker(d_s)$. This is particularly useful if $\ker(d_s)$ is finitely generated.

Continuing this, we can get a (full) projective resolution.

Definition 1.17

The *standard (or bar) resolution of* $\mathbb{Z}$, for any group G, is: Let

$$G^{(n)} = \{[g_1|\ldots|g_n] \mid g_i \in G\}$$

be the set of symbols. In particular, $G^{(0)} = \{[]\}$ Let

$$F_n = \mathbb{Z}G\{G^{(n)}\}$$

be the free module on $G^{(n)}$. The differential is given by

$$\begin{aligned} d_n[g_1|\ldots|g_n] = {} & g_1[g_2|\ldots|g_n] \\ & -[g_1g_2|g_3|\ldots|g_n] \\ & +[g_1|g_2g_3|\ldots|g_n] + \ldots \\ & +(-1)^{n-1}[g_1|\ldots|g_{n-1}g_n] \\ & +[g_1|\ldots|g_{n-1}] \end{aligned}$$

With this, $d_{n-1}d_n = 0$, and we have a chain complex

$$\cdots \longrightarrow F_n \longrightarrow F_{n-1} \longrightarrow \cdots \longrightarrow F_0 \xrightarrow{[]\mapsto 1} \mathbb{Z} \longrightarrow 0$$

Remark 1.18. The bar resolution corresponds to the standard resolution in algebraic topology. Consider $n+1$ tuples G^{n+1}, and form the free abelian group $\mathbb{Z}G^{n+1}$ under addition. G acts on G^{n+1} diagonally, so

$$g \cdot (h_0, \dots, h_n) = (gh_0, \dots, gh_n)$$

So $\mathbb{Z}G^{n+1}$ is a free $\mathbb{Z}G$-module, on the basis of $n+1$ tuples with $g_0 = 1$. Then we have a correspondence

$$[g_1|\dots|g_n] \mapsto (1, g_1, g_1g_2, \dots, g_1 \cdots g_n)$$

Note that removing the first entry gives $g_1(1, g_2, g_2g_3, \dots, g_2 \cdots g_n)$, and removing the seconc entry in the tuple gives $(1, g_1g_2, g_1g_2g_3, \dots, g_1 \cdots g_n)$.

Lemma 1.19. The bar resolution is exact.

Proof. We'll just consider the d_n as maps of abelian groups. F_n has basis $G \times G^{(n)}$ as a free abelian group, and $G \times G^{(n)}$ is the set

$$\{g_0[g_1|\dots|g_n] \mid g_i \in G\}$$

Define $\mathbb{Z}$-linear maps $s_n : F_n \to F_{n+1}$, such that

$$\mathrm{id}_{F_n} = \mathrm{d}_{n+1}s_n + s_{n-1}\mathrm{d}_n \tag{$*$}$$

given by

$$s_n(g_0[g_1|\dots|g_n]) = [g_0|\dots|g_n]$$

We can verify that $(*)$ holds on the basis, and so if $x \in \ker(\mathrm{d}_{n+1})$, then

$$x = \mathrm{id}(x) = \mathrm{d}_{n+1}s_n(x) + s_{n-1}\mathrm{d}_n(x) = \mathrm{d}_{n+1}s_n(x) \in \mathrm{im}(\mathrm{d}_{n+1})$$

That is, s_n is a chain homotopy from the identity to zero. □

Corollary 1.20. A finite group is of type FP_∞.

Proof. The bar resolution gives such a resolution. □

Lecture 3

Definition 1.21

Consider a projective resolution

$$\cdots \longrightarrow P_{n+1} \longrightarrow P_n \xrightarrow{d_n} \cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow \mathbb{Z} \longrightarrow 0$$

of $\mathbb{Z}$ by $\mathbb{Z}G$-modules. Let M be a (left) $\mathbb{Z}G$-module. Apply the functor $\mathrm{Hom}_G(\cdot, M)$, we get the *dual sequence*

$$\cdots \longleftarrow \mathrm{Hom}(P_{n+1}, M) \xleftarrow{\mathrm{d}^{n+1}} \mathrm{Hom}(P_n, M) \longleftarrow \cdots \longleftarrow \mathrm{Hom}(P_0, M) \tag{$\dagger$}$$

where $\mathrm{d}^n = \mathrm{d}_n^*$. Then the *$n$-th cohomology group* $\mathrm{H}^n(G, M)$ *with coefficients in* M *is:*

$$\mathrm{H}^n(G, M) = \frac{\ker(\mathrm{d}^{n+1})}{\mathrm{im}(\mathrm{d}^n)} \text{ for } n \geq 1$$
$$\mathrm{H}^0(G, M) = \ker(\mathrm{d}^1)$$

Remark 1.22. We have dropped the $\mathbb{Z}$-term in $(\dagger)$.

Also, these cohomology groups are the homology groups of the chain complex

$$C_n = \mathrm{Hom}_G(P_{-n}, M)$$

with $-\infty < n \leq 0$.

Later, we will see that these cohomology groups are independent of the choice of resolution.

Example 1.23

In our prototypical example, $G = \langle t \rangle$ infinite cyclic, we have a projective resolution

$$0 \longrightarrow \mathbb{Z}G \xrightarrow{\mathrm{d}} \mathbb{Z}G \longrightarrow \mathbb{Z} \longrightarrow 0$$

where $\mathrm{d}(x) = x(t-1)$. For $\phi \in \mathrm{Hom}_G(\mathbb{Z}G, M)$, $x \in \mathbb{Z}G$,

$$\mathrm{d}^1(\phi)(x) = \phi(\mathrm{d}_1(x)) = \phi(x(t-1))$$

Moreover, recall $\mathrm{Hom}_G(\mathbb{Z}G, M) \cong M$, where $i(\phi) = \phi(1)$. In particular,

$$\mathrm{d}^1(\phi) \mapsto \mathrm{d}^1\phi(1) = \phi(t-1) = (t-1)\phi(1) = (t-1)i(\phi)$$

In this case, we have

$$0 \longleftarrow M \xleftarrow[(t-1)\cdot]{} M$$

and so

$$\mathrm{H}^0(G, M) = \{m \in M \mid tm = m\} = M^G$$
$$\mathrm{H}^1(G, M) = \frac{M}{(t-1)M} = M_G$$

and $\mathrm{H}^n(G, M) = 0$ otherwise. Here, M^G is the group of *invariants*, the largest submodule with trivial G-action. Conversely, M_G is the *coinvariants*, the largest quotient module with trivial G-action.

Remark 1.24. $\mathrm{H}^0(M, G) = M^G$ is generally true. But $\mathrm{H}^1(M, G) = M_G$ is special behaviour of $G = \langle t \rangle$.

For any group of type FP, we have that $\mathrm{H}^n(G, M) = 0$ for large enough n.

Definition 1.25

G is of *cohomological dimension m over $\mathbb{Z}$* if there exists an $\mathbb{Z}G$-module M, with

$$\mathrm{H}^m(G, M) \neq 0$$

and $\mathrm{H}^n(G, M') = 0$ for any $\mathbb{Z}G$-module M', and $n > m$.

Note that for all G, $\mathrm{H}^0(G, \mathbb{Z}) = \mathbb{Z} \neq 0$.

Example 1.26

$G = \langle t \rangle$ infinite cyclic is of cohomological dimension 1 (over $\mathbb{Z}$).

An exercise is to show that if G is a free group of finite rank, then it is also of cohomological dimension 1. To see this, the Cayley graph is a tree and we can construct a resultion using this.

Remark 1.27. The converse is true. A finitely generated group of cohomological dimension 1 is free. In fact, this is true in general. See [Stallings, 1968], Swan, 1969.

Now consider the bar resolution in our definition of cohomology. Note that

$$\mathrm{Hom}_G(\mathbb{Z}G\{G^{(n)}\}, M) \cong C^n(G, M) = \{\text{functions } G^{(n)} \to M\}$$

This is the same as function $G^n \to M$. We also have that

$$C^0(G, M) = \{\text{functions } [] \to M\} = M$$

Definition 1.28

The group of *n-cochains* of G with coefficients in M is $C^n(G, M)$. The *n-coboundary map*

$$\mathrm{d}^n : C^{n-1}(G, M) \to C^n(G, M)$$

is dual to d_n in the bar resolution. For $\phi \in C^{n-1}(G, M)$,

$$\begin{aligned}(\mathrm{d}^n\phi)(g_1, \ldots, g_n) = {} & g_1\phi(g_2, \ldots, g_n) \\ & - \phi(g_1g_2, \ldots, g_n) \\ & + \phi(g_1, g_2g_3, \ldots, g_n) \\ & + \ldots \\ & + (-1)^{n-1}\phi(g_1, g_2, \ldots, g_{n-1}g_n) \\ & + (-1)^n\phi(g_1, g_2, \ldots, g_{n-1})\end{aligned}$$

The group of *n-cocycles* is $Z^n(G, M) = \ker(\mathrm{d}^{n+1}) \subseteq C^n(G, M)$, and the *n-coboundaries* is $B^n(G, M) = \mathrm{im}(\mathrm{d}^n) \subseteq C^n(G, M)$.

Thus,

$$\mathrm{H}^n(G, M) = \frac{Z^n(G, M)}{B^n(G, M)}$$

Corollary 1.29. For any group G, $\mathrm{H}^0(G, M) = M^G$ is the invariants.

Definition 1.30

A *derivation of G with coefficients in M* is a function $\phi : G \to M$, such that

$$\phi(gh) = g\phi(h) + \phi(g)$$

Note that $Z^1(G, M)$ is the set of derivations. Also note that these are also called 'crossed homomorphisms'.

An *inner derivation* is ϕ of the form

$$\phi(g) = gm - m$$

for a fixed $m \in M$.

Corollary 1.31.

$$\mathrm{H}^1(G, M) = \frac{\{\text{derivations } G \to M\}}{\{\text{inner derivations} G \to M\}}$$

In particular, if M is a trivial $\mathbb{Z}G$-module, then

$$\mathrm{H}^1(G, M) = \mathrm{Hom}_{\mathrm{Group}}(G, M)$$

We will return to considering homology arising from different resolutions.

Definition 1.32

Let $(A_n, \alpha_n), (B_n, \beta_n)$ be chain complexes of $\mathbb{Z}G$-modules, a *chain map* $f = (f_n)$ are $\mathbb{Z}G$-maps $f_n : A_n \to B_n$,

such that

$$\begin{array}{ccc} A_n & \xrightarrow{\alpha_n} & A_{n-1} \\ {\scriptstyle f_n}\downarrow & & \downarrow{\scriptstyle f_{n-1}} \\ B_n & \xrightarrow[\beta_n]{} & B_{n-1} \end{array}$$

commutes.

Lemma 1.33. Given a chain map (f_n), it induces a map on homology groups

$$f_* : \mathrm{H}_n(A_\bullet) \to \mathrm{H}_n(B_\bullet)$$

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